MATH441/MATH541
Applied Mathematics II: Complex Variables
Lecture slides 2: Polar form and roots of complex numbers

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Lecture outline

1. Examples of calculations with complex numbers
2. Geometry of complex plane
3. Trigonometric form of complex numbers
4. Triangle inequality and related formulas
5. Geometry of complex multiplication and division
6. Powers and roots of complex numbers
7. Simple geometric objects: lines, half planes

Example: Numerical calculations with complex numbers

Example

Compute complex numbers:

1. $(3-4i)(6+2i) = 18 - 6i - 24i + 8 = 26 - 18i$
2. $\frac{2+i}{4-7i}\left(\frac{4+7i}{4+7i}\right) = \frac{8+14i+4i-7}{16+49} = \frac{1+18i}{65} = \frac{1}{65} + \frac{18}{65}i$
3. $(17-6i)\overline{(-4-12i)} = (17-6i)(-4+12i) = -68 + 204i + 24i + 72 = 4 + 228i$
4. $\left(\frac{-6+2i}{1-8i}\right)^2$

   1st step: $\left(\frac{-6+2i}{1-8i}\right)\left(\frac{1+8i}{1+8i}\right) = \frac{-6-48i+2i-16}{1+64} = \frac{-22-46i}{65}$

   2nd step: $(\frac{-22-46i}{65})^2 = \frac{-1632}{4225} + \frac{2024}{4225}i$

Example: Power of i

Example

Prove that for any positive integer n
$i^{4n} = 1, \qquad i^{4n+1} = i \qquad i^{4n+2} = -1 \qquad i^{4n+3} = -i$

- $i^{4n} = (i^4)^n = 1^n = 1.$
- $i^{4n+1} = (i^{4n})i = 1i = i.$
- $i^{4n+2} = (i^{4n})i^2 = -i^{4n} = -1.$
- $i^{4n+3} = (i^{4n})i^3 = (i^{4n})(-i) = -i.$

Example: Prove of a simple statement. Part 1.

Example

Prove that z is real or pure imaginary $\iff z^2 = \bar{z}^2$.

Suppose z is real.

Then if $z = x + iy$, $y = 0$.

$$\begin{aligned} z^2 = (x+iy)^2 = (x+iy)(x+iy) \\ &= x^2 + 2xyi - y^2 \\ &= x^2 \\ &= x^2 - 2xyi + y^2 \\ &= (x-iy)(x-iy) \\ &= (x-iy)^2 \\ &= \bar{z}^2 \end{aligned}$$

Example: Prove of a simple statement. Part 2.

Example

Suppose z is pure imaginary.
Then if $z = x + iy$, $x = 0$.
So,
$$\begin{aligned} z^2 = (x+iy)^2 &= (x+iy)(x+iy) \\ &= x^2 + 2xyi - y^2 \\ &= -y^2 \\ &= x^2 - 2xyi - y^2 \\ &= (x-iy)(x-iy) \\ &= (x-iy)^2 \\ &= \overline{z}^2 \end{aligned}$$

Now, suppose $z^2 = \overline{z}^2$

then,
$$\begin{aligned} (x+iy)^2 &= (x-iy)^2 \\ (x+iy)(x+iy) &= (x-iy)(x-iy) \\ x^2 + 2xyi - y^2 &= x^2 - 2xyi - y^2 \\ 2xyi &= -2xyi \\ 2xy &= -2xy \end{aligned}$$

The only solutions which satisfy this are either $x = 0$ or $y = 0$.
So then, z is either real or pure imaginary.

Geometric representation of complex numbers

Complex numbers can be represented geometrically as points in the plane, a fact which is not only useful but virtually indispensable. Introducing a rectangular coordinate system in the plane, we can identify the complex number $z = x + iy$ with the point $P = (x, y)$. In this way we establish a one-to-one correspondence between the set of all complex numbers and the set of all points in the plane. Clearly, under this mapping, the set of all real numbers corresponds to the x-axis (hence called the *real axis*) and the set of all purely imaginary numbers corresponds to the y-axis (hence called the *imaginary numbers*), while the set of all imaginary numbers corresponds to all points which do not lie on the real axis. Moreover, the complex number 0 corresponds to the point of intersection of the x-axis and y-axis, i.e. the origin of coordinates. The plane whose points represent the complex numbers is called the *complex plane*, or the *z-plane*, *w-plane*, ..., depending on the letter z, w, ... used to denote a generic complex number. With the understanding that such a complex plane has been constructed, the terms "complex number $x + iy$" and "point $x + iy$" can be used interchangeably.

Polar form of complex numbers 1

Another entirely equivalent way of representing the complex number $z = x + iy$ is to use the vector $\overrightarrow{OP}$ joining the origin O of the complex plane to the point $P = (x, y)$, instead of using the point P itself. The length of the vector $\overrightarrow{OP}$ is called the *modulus* or *absolute value* of the complex number z, and is denoted by $|z|$. The angle between the positive real axis and the vector $\overrightarrow{OP}$, more exactly, the angle through which the positive real axis must be rotated to cause it to have the same direction as $\overrightarrow{OP}$ (considered positive if the rotation is counterclockwise and negative otherwise), is called the *argument* of the complex number z, and is denoted by $Arg\, z$. In other words, $|z|$ and $Arg\, z$ are the polar coordinates r and Φ of the point with rectangular coordinates x and y, i.e.

$$\begin{aligned} x &= Re\, z = r\cos\Phi, \\ y &= Im\, z = r\sin\Phi. \end{aligned}$$

It follows at once that

$$z = x + iy = r(\cos\Phi + i\sin\Phi), \tag{1}$$

where (1) is called the *trigonometric form* of the complex number z.

Figure: Algebraic and polar forms of complex number

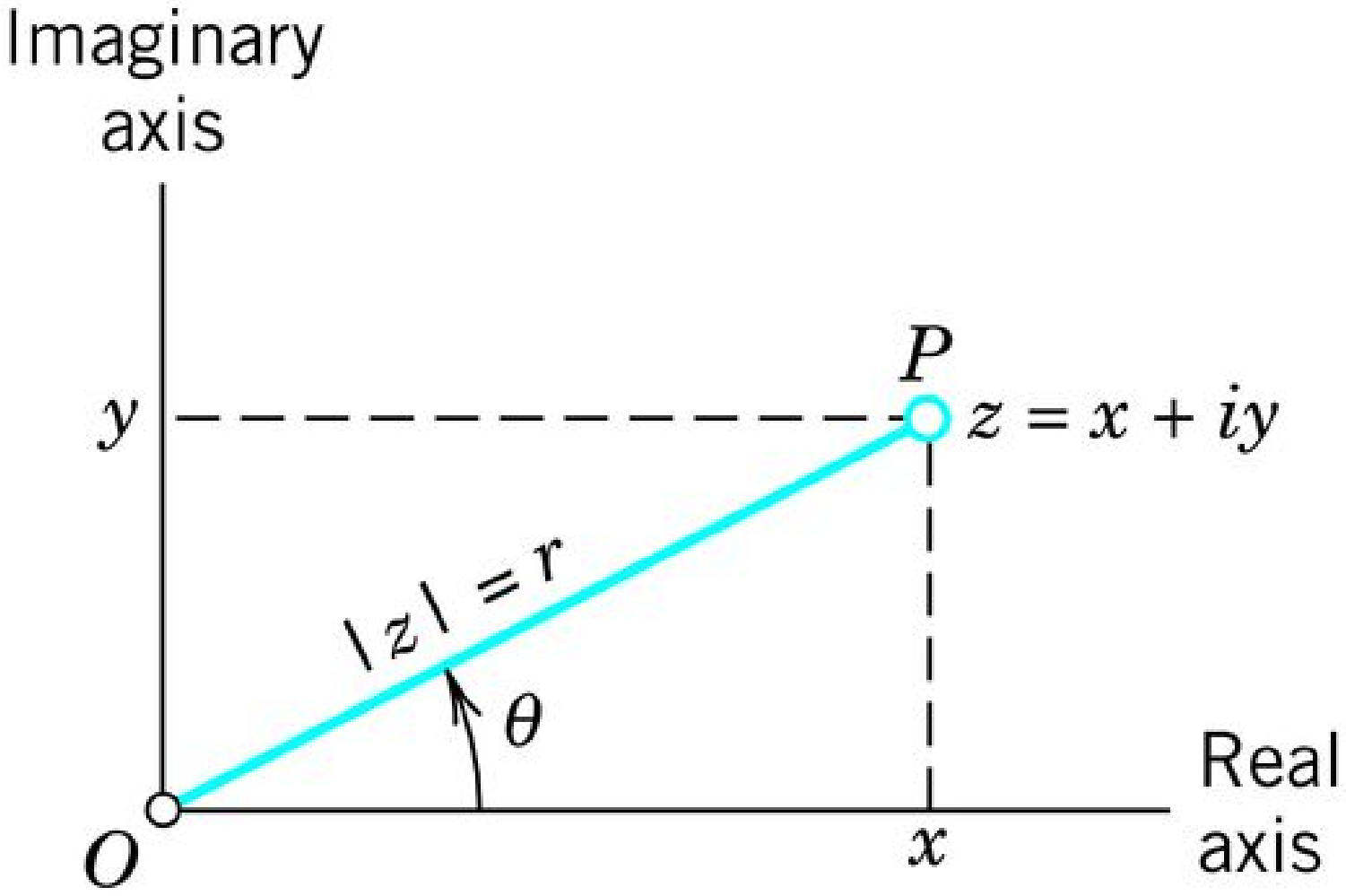


Figure: Algebraic and polar forms of complex number

Polar form of complex numbers: *Arg z*

Clearly, the quantity *Arg z* is defined only to within an integral multiple of 2π. However, there is one and only one value of *Arg z* say ϕ, which satisfies the inequality

$$-\pi < \phi \leq \pi,$$

and we shall call ϕ the *principal value* of the argument z, written *arg z*. The relation between *Arg z* and *arg z* is given by

$$Arg\, z = arg\, z + 2n\pi,$$

where n ranges over all the integers $0, \pm 1, \pm 2, \ldots$. It is an immediate consequences of our definitions that

$$|z| = \sqrt{x^2 + y^2}$$

and

$$\tan(arg\, z) = \frac{y}{x}. \tag{2}$$

Polar form of complex numbers: computing *arg z*

Some care is required in inverting (2), since the arc tangent of a real number x, written $Arc\ \tan x$, is only defined to within an integral multiple of π. However, there is one and only one value of $Arc\ \tan x$, say α, which satisfies the inequality

$$-\frac{\pi}{2} < \alpha \leq \frac{\pi}{2},$$

and we shall call α the *principal value* of the arc tangent of x, written $arc\ \tan x$. We can now invert the relation (2), obtaining

$$arg\ z = \begin{cases} arc\ \tan \frac{y}{x} & \text{if } x > 0, \\ arc\ \tan \frac{y}{x} + \pi & \text{if } x < 0,\ y \geq 0, \\ arc\ \tan \frac{y}{x} - \pi & \text{if } x < 0,\ y < 0. \end{cases}$$

Moreover, when y/x becomes infinite, we clearly have

$$arg\ z = \begin{cases} \frac{\pi}{2} & \text{if } x = 0,\ y > 0, \\ -\frac{\pi}{2} & \text{if } x = 0,\ y < 0, \end{cases}$$

while the case $z = 0$ (i.e. $x = y = 0$) is indeterminate.

Example: finding *arg z* of $-1+4i$. Part 1

Find all of the arguments of $z = -1+4i$

z is in the second quadrant:

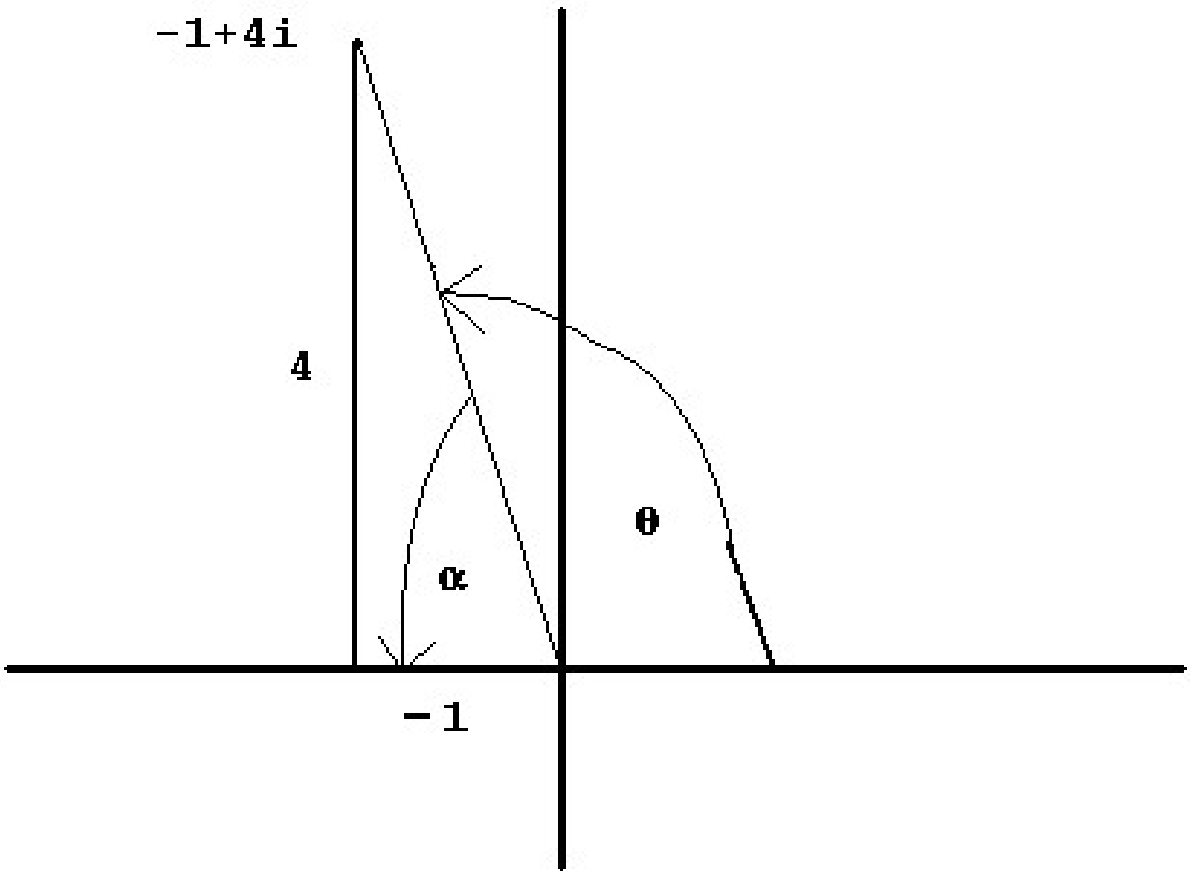


Figure: $-1+4i$

Example: finding *arg z* of $-1+4i$. Part 2

With the α shown, $\tan\alpha = 4$, so $\alpha = \tan^{-1} 4$.
The angle θ is one argument of $-1+4i$, and

$$\theta = \pi - \tan^{-1} 4.$$

Hence

$$(-1+4i) = \left\{\pi - \tan^{-1} 4 + 2\pi k \middle| \ k \text{ an integer}\right\}$$

Complex conjugates in trigonometric form

The complex numbers $x + iy$ and $x - iy$ are said to be *conjugate complex numbers* or *complex conjugates* (of each other). If one of these numbers is denoted by z, the other is denoted by $\bar{z}$. Obviously, the points z and $\bar{z}$ are symmetric with respect to the real axis, and

$$\begin{aligned} |\bar{z}| &= |z|, \\ \overline{(\bar{z})} &= z. \end{aligned}$$

Moreover,

$$arg\, z = -arg\, z,$$

unless z is a negative number, in which case

$$arg\, z = arg\, \bar{z} = \pi.$$

Triangle inequality and related formulas. Part 1

Geometrically, addition of two complex numbers z_1 and z_2 corresponds to addition of the vectors representing them. Similarly the difference $z_1 - z_2$ is represented by the vector joining the point z_2 to the point z_1. It follows that $\rho(z_1, z_2)$, the distance between the two points z_1 and z_2, equals the modulus of the complex number $z_1 - z_2$:

$$\rho(z_1, z_2) = |z_1 - z_2|.$$

Then the familiar fact that the length of one side of a triangle cannot exceed the sum of the lengths of the other two sides implies the following two inequalities involving complex numbers:

$$|z_1 + z_2| \leqslant |z_1| + |z_2|, \tag{3}$$

$$|z_1 - z_2| = ||z_1| - |z_2||. \tag{4}$$

The relation (3) can be generalized immediately to the case of n complex numbers:

$$|z_1 + z_2 + \ldots + z_n| \leqslant |z_1| + |z_2| + \ldots + +|z_n|. \tag{5}$$

Figure: Triangle inequality

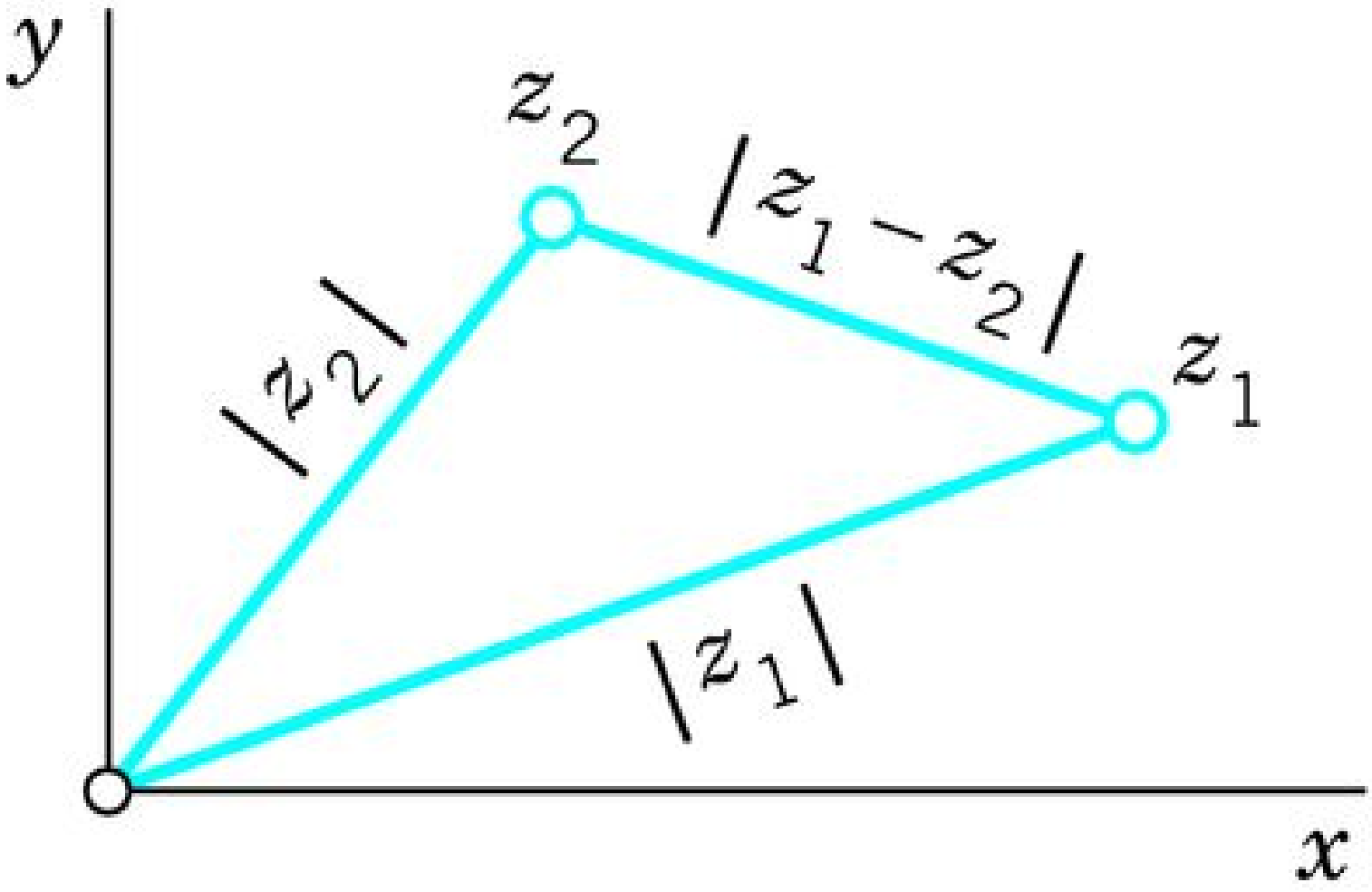


Figure: Triangle inequality

Triangle inequality and related formulas. Part 2

In (3), (4) and (5), equality occurs if and only if all the (nonzero) complex numbers have the same argument. We note that if $z = x + iy$, then

$$z\bar{z} = x^2 + y^2 = |z|^2,$$

and hence

$$|z| = \sqrt{x^2 + y^2},$$

and

$$|z_1 + \cdots + z_n| = \sqrt{(z_1 + \cdots + z_n)(\bar{z}_1 + \cdots + \bar{z}_n)}.$$

Summary of simple formulas

Theorem

$\forall\ z, w \in \mathbb{C}$ *we have*

1. $\overline{(\bar{z})} = z$
2. $\overline{z+w} = \bar{z} + \bar{w}$
3. $\overline{zw} = \bar{z}\bar{w}$
4. $|\bar{z}| = |z|$
5. $|zw| = |z|\,|w|$
6. *Real part of* $z = Re(z) = \frac{1}{2}(z + \bar{z})$ *and Imaginary part of* $z = Im(z) = \frac{1}{2i}(z - \bar{z})$
7. $|z| \geq 0$ *and* $|z| = 0 \Leftrightarrow z = 0$.
8. $z\bar{z} = |z|^2$

Summary of simple inequalities

Theorem

Inequalities: $\forall z, w \in \mathbb{C}$
1. $|Re(z)| \leq |z|$ *2.* $|Im(z)| \leq |z|$
3. $|z+w| \leq |z| + |w|$ *4.* $|\,|z| - |w|\,| \leq |z+w|$

Proof of (3):
$0 \leq |z+w|^2 = (z+w)\overline{(z+w)} = (z+w)(\bar{z}+\bar{w}) = z\bar{z} + z\bar{w} + w\bar{z} + w\bar{w}$

$= |z|^2 + z\bar{w} + \overline{(z\bar{w})} + |w|^2 = |z|^2 + 2Re(z\bar{w}) + |w|^2$

$\leq |z|^2 + 2\,|z\bar{w}| + |w|^2 = |z|^2 + 2\,|z|\,|w| + |w|^2 = (|z| + |w|)^2$
Proof of (4): $|z| = |z+w-w| \leq |z+w| + |w|$
Hence $|z| - |w| \leq |z+w|$ Now interchanging z and w
$\Longrightarrow |w| - |z| \leq |w+z|$. Hence

$$-|z+w| \leq |z| - |w| \leq |z+w|$$

which is equivalent to 4.

Geometry of complex numbers multiplication. Part 1

To give a geometric interpretation of multiplication and division of two complex numbers z_1 and z_2, we first write z_1 and z_2 in trigonometric form, i.e.

$$\begin{aligned} z_1 &= r_1(\cos\Phi_1 + i\sin\Phi_1), \\ z_2 &= r_2(\cos\Phi_2 + i\sin\Phi_2), \end{aligned}$$

where

$$\begin{aligned} r_1 &= |z_1|, \ \Phi_1 = Arg\, z_1, \\ r_2 &= |z_2|, \ \Phi_2 = Arg\, z_2. \end{aligned}$$

Geometry of complex numbers multiplication. Part 2

Then,

$$\begin{aligned} z_1 z_2 &= r_1 r_2 \left(\left(\cos \Phi_1 \cos \Phi_2 - \sin \Phi_1 \sin \Phi_2 \right) \right) + i \left(\sin \Phi_1 \cos \Phi_2 + \cos \Phi_1 \sin \Phi_2 \right) \right) \\ &= r_1 r_2 \left(\cos \left(\Phi_1 + \Phi_2 \right) + i \sin \left(\Phi_1 + \Phi_2 \right) \right), \end{aligned} \tag{6}$$

and therefore

$$\begin{aligned} |z_1 z_2| &= |z_1| \, |z_2|, \\ Arg\, z_1 z_2 &= Arg\, z_1 + Arg\, z_2. \end{aligned}$$

The last formula means that the set of values of $Arg\, z_1 z_2$ is obtained by forming all possible sums of a value of $Arg\, z_1$ and a value of $Arg\, z_2$. Thus the geometric interpretation of multiplication of z_1 by z_2 ($z_1 \neq 0$, $z_2 \neq 0$) is the following: Draw the vector z_1 (with initial point at the origin), multiply its length by the factor $|z_2|$, and rotate the resulting vector counterclockwise through the angle $Arg\, z_2$.

Geometric interpretation of division

Similarly, if $z_2 \neq 0$, it is easy verified that

$$\left| \frac{z_1}{z_2} \right| = \frac{|z_1|}{|z_2|}, \tag{7}$$

$$Arg \frac{z_1}{z_2} = Arg\, z_1 - Arg\, z_2, \tag{8}$$

with a corresponding geometric interpretation of division. It follows from (8) that $Arg\,(z_1/z_2)$ is the angle between the vectors z_2 and z_1, measured from z_2 to z_1 in the counterclockwise direction.

Powers complex numbers. Part 1

Let n be a positive integer and z a complex number. Then by the nth *power* of z, written z^n, we mean z multiplied by itself n times, just as in the case of real numbers. If $z = 0$, we obviously have $z^n = 0$. If $z \neq 0$, then, setting

$$z = r(\cos \Phi + i \sin \Phi) \tag{9}$$

and repeatedly applying the formula for the multiplication of two complex numbers in trigonometric form, we find that

$$z^n = r^n(\cos n\Phi + i \sin n\Phi). \tag{10}$$

When $r = 1$, (9) and (10) together imply the following formula, known as *De Moivre's theorem*:

$$(\cos \Phi + i \sin \Phi)^n = \cos n\Phi + i \sin n\Phi. \tag{11}$$

Powers complex numbers. Part 2

Using the binomial theorem to expand the left-hand side of (11) and equating real and imaginary parts of the resulting equation, we can express $\cos n\Phi$ or $\sin n\Phi$ entirely in terms of powers of $\cos\Phi$ and $\sin\Phi$. For the same algebraic reasons as in the case of real numbers, we define

$$\begin{aligned} z^0 &= 1, \\ z^{-n} &= \frac{1}{z^n}. \end{aligned}$$

Then formulas (10) and (11) are valid for all integers $n = 0, \pm 1, \pm 2, \ldots$.

Roots of complex numbers. Part 1

Next we consider the problem of extracting the root of a complex number. Again let n be a positive integer and z a complex number. Then by the nth *root* of z, written $\sqrt[n]{z}$ or $z^{1/n}$, we mean any complex number ζ satisfying the equation

$$\zeta^n = z. \tag{12}$$

If $z = 0$, we obviously have $\zeta = 0$. If $z \neq 0$, then, setting

$$\begin{aligned} \zeta &= \rho(\cos\Theta + i\sin\Theta), \\ z &= r(\cos\Phi + i\sin\Phi), \end{aligned}$$

we find that

$$\rho^n(\cos n\Theta + i\sin n\Theta) = r(\cos\Phi + i\sin\Phi). \tag{13}$$

Roots of complex numbers. Part 2

It follows that (by $\sqrt[n]{a}$, where $a > 0$ we always mean the unique *positive* root of a)

$$\begin{aligned} \rho^n &= r, \\ \rho &= \sqrt[n]{r} = \sqrt[n]{|z|} \end{aligned}$$

and

$$n\Theta = \Phi + 2m\pi \quad (m = 0, \pm 1, \pm 2, \ldots), \tag{14}$$

i.e.

$$\begin{aligned} n\Theta &= Arg\, z, \\ \Theta &= \frac{Arg\, z}{n}. \end{aligned}$$

Therefore we have

$$\sqrt[n]{z} = \sqrt[n]{|z|}\left(\cos\frac{Arg\, z}{n} + i\sin\frac{Arg\, z}{n}\right). \tag{15}$$

Roots of complex numbers. Part 3

It is clear from (15) that values of $Arg\, z$ which differ by $2k\pi$, where k is an integer not divisible by n, correspond to different values of $\sqrt[n]{z}$.
Accordingly, when $Arg\, z$ runs through the values

$$arg\, z,\ arg\, z + 2\pi, \ldots,\ arg\, z + 2(n-1)\pi, \tag{16}$$

we obtain n different values of $\sqrt[n]{z}$. Since any other value of $Arg\, z$ differs from one of the numbers (16) by a multiple of $2n\pi$, these n values of $\sqrt[n]{z}$ exhaust all the possibilities. Thus the nth root of z has n different values (if $z \neq 0$), which are all implicit in the formula (15). The value of $\sqrt[n]{z}$ equal to

$$\sqrt[n]{|z|}\left(\cos\frac{arg\, z}{n} + i\sin\frac{arg\, z}{n}\right). \tag{17}$$

is called the *principal value* of $\sqrt[n]{z}$, since when z is a positive real number, since when z is a positive real number, the principal value of $\sqrt[n]{z}$ coincides with the *positive* nth root of z.

Roots of complex numbers. Part 4

It follows from the formula

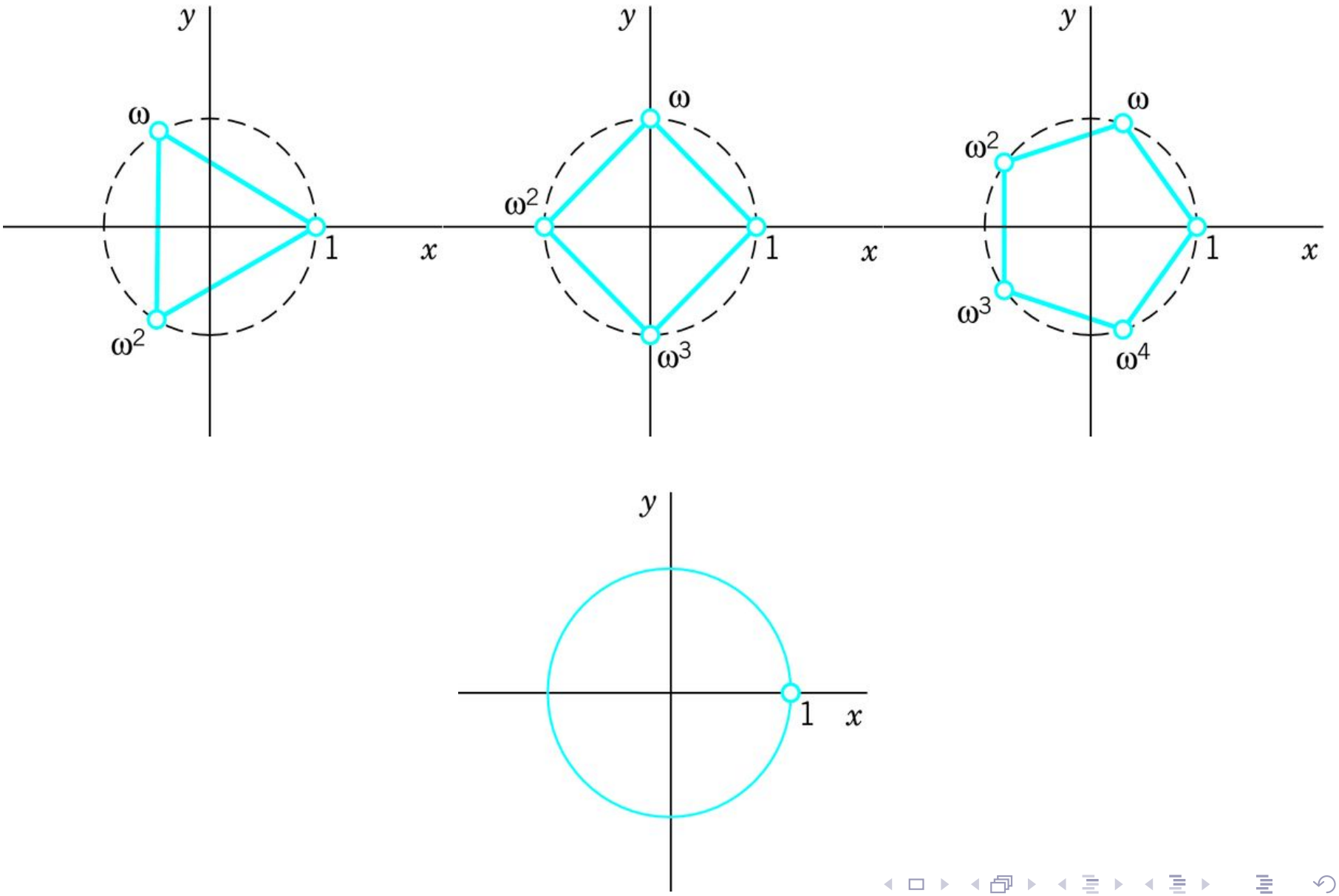
$$Arg\, z = arg\, z + 2m\pi = arg\, z + Arg\, 1 \tag{18}$$

that

$$\begin{aligned} \sqrt[n]{z} &= \sqrt[n]{|z|}\left(\cos\frac{arg\, z + Arg\, 1}{n} + i\sin\frac{arg\, z + Arg\, 1}{n}\right) \\ &= \sqrt[n]{|z|}\left(\cos\frac{arg\, z}{n} + i\sin\frac{arg\, z}{n}\right)\left(\cos\frac{Arg\, 1}{n} + i\sin\frac{Arg\, 1}{n}\right), \end{aligned} \tag{19}$$

i.e., all the values of $\sqrt[n]{z}$ can be obtained from the principal value by multiplying the latter by different nth roots of unity.

Figure: Roots of 1



Example: $\sqrt[6]{1}$ Sixth roots of unity.

Example

Find the sixth roots of unity.

$$z^{\frac{1}{n}} \text{ when } z = 1 \text{ is} \qquad \cos\left(\frac{2\pi k}{n}\right) + i\sin\left(\frac{2\pi k}{n}\right)$$

$$\begin{aligned}
&\text{root 1:} && 1 \\
&\text{root 2:} && \cos\left(\tfrac{\pi}{3}\right) + i\sin\left(\tfrac{\pi}{3}\right) = \tfrac{1}{2} + \tfrac{1}{2}i\sqrt{3} \\
&\text{root 3:} && \cos\left(\tfrac{2\pi}{3}\right) + i\sin\left(\tfrac{2\pi}{3}\right) = -\tfrac{1}{2} + \tfrac{1}{2}i\sqrt{3} \\
&\text{root 4:} && \cos(\pi) + i\sin(\pi) = -1 \\
&\text{root 5:} && \cos\left(\tfrac{4\pi}{3}\right) + i\sin\left(\tfrac{4\pi}{3}\right) = -\tfrac{1}{2} - \tfrac{1}{2}i\sqrt{3} \\
&\text{root 6:} && \cos\left(\tfrac{5\pi}{3}\right) + i\sin\left(\tfrac{5\pi}{3}\right) = \tfrac{1}{2} - \tfrac{1}{2}i\sqrt{3}
\end{aligned}$$

Example: the square roots of $-15-8i$. Part 1

Example

Find the square roots of $-15-8i$ by using the standard formula

- Step 1. $|-15-8i| = 17$
- Step 2. $\arctan(\frac{8}{15}) = .48996$,

  $$Arg(-15-8i) = \arctan\frac{8}{15} - \pi$$
- Step 3. The first root is

$$w = \sqrt{17}[\cos(\frac{\arctan\frac{8}{15}-\pi}{2}) + i\sin(\frac{\arctan\frac{8}{15}-\pi}{2})]$$

The second root is

$$w = \sqrt{17}[\cos(\frac{\arctan\frac{8}{15}-\pi}{2}+\pi) + i\sin(\frac{\arctan\frac{8}{15}-\pi}{2}+\pi)]$$

Example: the square roots of $-15-8i$. Part 2.

Example

Find the square roots of $-15-8i$ by assuming that $p+iq$ is the required root and finding p and q.

Suppose now that $p+iq$ is a root of $-15-8i$, where p and q are real. Then
$(p+iq)^2 = p^2+2ipq-q^2 = -15-8i$.
Thus $2pq=-8$ or $q=-\frac{4}{p}$ and $p^2-q^2=-15$.
Hence

$$p^2-\frac{16}{p^2}=-15,$$

$$p^4+15p^2-16=0$$

Solution is: $\{p=1\}, \{p=-1\}, \{p=4i\}, \{p=-4i\}$.
We reject the last two solutions since p is real.
Thus $p=\pm 1, q=\mp 4$ so the roots are $1-4i$ and $-1+4i$.

Rational power of complex numbers.

Finally we define an arbitrary rational power of a complex number by the formula

$$z^{m/n} = \left(\sqrt[n]{z}\right)^m,$$

where m and $n > 0$ are relatively prime integers. Then we have

$$\begin{aligned} z^{m/n} &= \left[\sqrt[n]{|z|}\left(\cos\frac{Arg\ z}{n} + i\sin\frac{Arg\ z}{n}\right)\right]^m \\ &= \left(\sqrt[n]{|z|}\right)^m\left(\cos\frac{m\,Arg\ z}{n} + i\sin\frac{m\,Arg\ z}{n}\right) \\ &= |z|^{m/n}\left(\cos\frac{m\,Arg\ z}{n} + i\sin\frac{m\,Arg\ z}{n}\right), \end{aligned} \tag{20}$$

where $|z|^{m/n}$ is defined as the positive number $\left(\sqrt[n]{|z|}\right)^m = \sqrt[n]{|z|^m}$.

Lines and half planes in the complex plane 1

Let L denote a straight line in $\mathbb{C}$. From elementary analytic geometry, L is determined by a point in L and a direction vector. Thus if a is any point in L and $\mathbf{b}$ is its direction vector then

$$L = \{z = a + t\mathbf{b} \mid -\infty < t < \infty\}.$$

Since $b \neq 0$ this gives, for z in L,

$$Im\left(\frac{z-a}{\mathbf{b}}\right) = 0.$$

In fact if z is such that

$$0 = Im\left(\frac{z-a}{\mathbf{b}}\right)$$

then

$$t = \left(\frac{z-a}{\mathbf{b}}\right)$$

implies that $z = a + t\mathbf{b}, -\infty < t < \infty$.

Lines and half planes in the complex plane 2

That is

$$L = \left\{ z \mid Im\left(\frac{z-a}{\mathbf{b}}\right) = 0 \right\}. \tag{21}$$

What is the locus of each of the sets

$$\left\{ z \mid Im\left(\frac{z-a}{\mathbf{b}}\right) > 0 \right\},$$

$$\left\{ z \mid Im\left(\frac{z-a}{\mathbf{b}}\right) < 0 \right\}?$$

As a first step in answering this question, observe that since $\mathbf{b}$ is a direction we may assume $|\mathbf{b}| = 1$. For the moment, let us consider the case where $a = 0$, and put $H_0 = \{z \mid Im\,(z/\mathbf{b}) > 0\}$, $\mathbf{b} = \cos\beta + i\sin\beta$. If $z = r(\cos\theta + i\sin\theta)$ then $z/b = r(\cos(\theta-\beta) + i\sin(\theta-\beta))$. Thus, z is in H_0 if and only if $\sin(\theta-\beta) > 0$; that is, when $\beta < \theta < \pi + \beta$. Hence H_0 is the half plane lying to the left of the line L if we are "walking along L in the direction of b".

Lines and half planes in the complex plane 3

If we put

$$H_a = \left\{ z \mid Im\left(\frac{z-a}{\mathbf{b}}\right) > 0 \right\},$$

then it is easy to see that $H_a = a + H_0 = \{a + w \mid w \in H_0\}$; that is, H_a is the translation of H_0 by A. Hence, H_a is the half plane lying to the left of L. Similarly,

$$K_a = \left\{ z \mid Im\left(\frac{z-a}{\mathbf{b}}\right) < 0 \right\},$$

is the half plane on the right of L.

Example

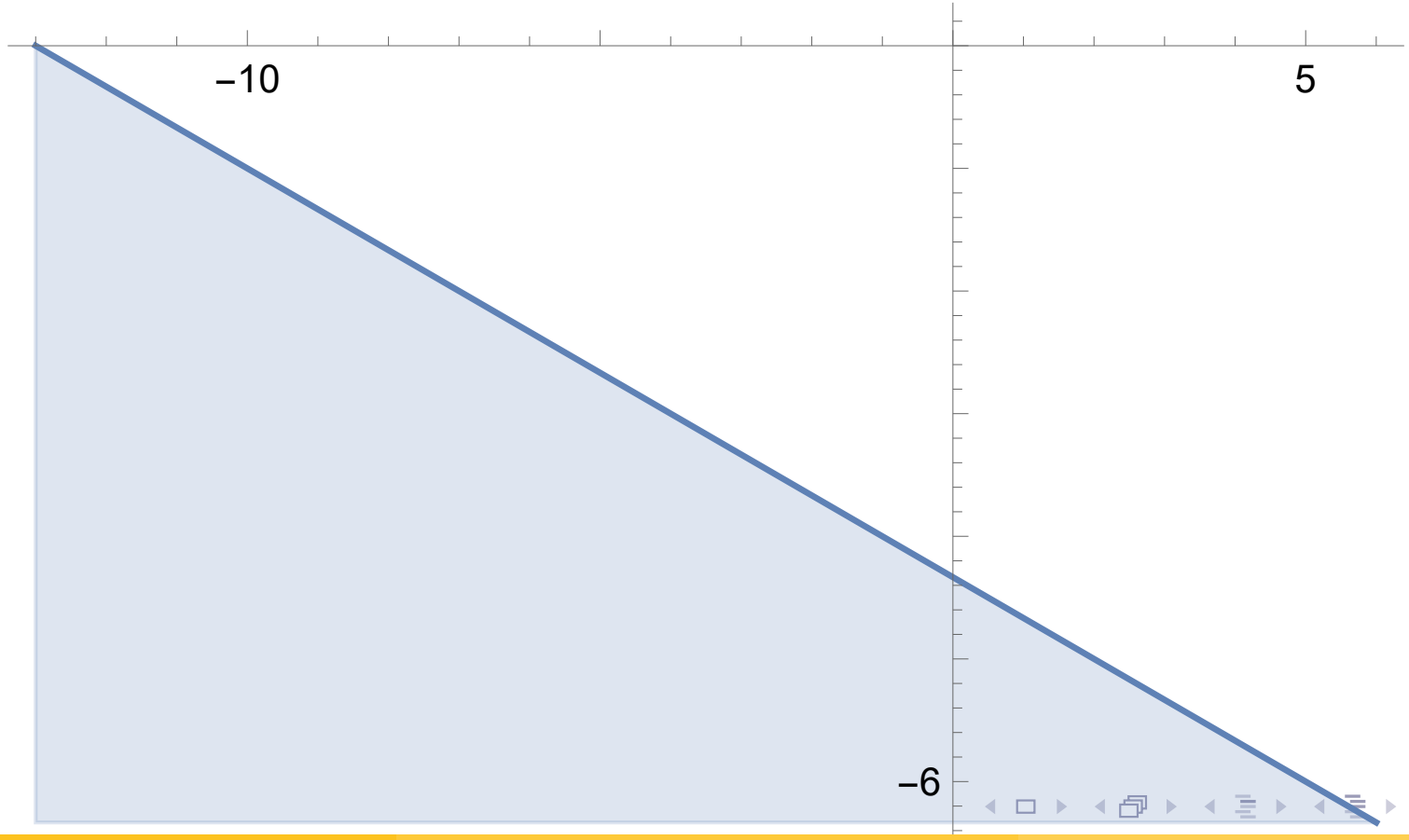
Sketch the locus of points satisfying $|z+6i| \leq |z-1+3i|$

$$x^2 + (y+6)^2 \leq (x-1)^2 + (y+3)^2$$

$x^2 + (y+6)^2 = x^2 + y^2 + 12y + 36$ and

$(x-1)^2 + (y+3)^2 = x^2 - 2x + 10 + y^2 + 6y$

$x^2 + y^2 + 12y + 36 \leq x^2 - 2x + 10 + y^2 + 6y$, Solution is: $\left\{y \leq -\frac{13}{3} - \frac{1}{3}x\right\}$



Thank you